What happens when predictions are disconfirmed? Disentangling the underlying processes by using an implicit memory test

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Previous Research

Predictions involve top-down **pre-activations** of the expected word, as well as of its semantic and form-related neighbors[1,2,3]

→ What happens with these pre-activated representations in memory when **predictions** are **disconfirmed**?

Previous studies used **explicit recognition memory tests**

- Lingering of expected & semantic words elicited false remembering [4,5,6]
- Possible suppression of form-related words resulted in no/reduced false remembering? [4,7]

But: Effects unlikely to be driven by prediction [4]

- Expected & semantic words related to the sentence context
- People likely used the sentence context to jog their memory

→ Are there **downstream prediction effects** when using an **implicit memory test**?

Study Design

1. Self-Paced Reading

38 experimental sentences (Cloze Probability: 0,59 - 1)

→ *Da Anne Angst vor Spinnen hat, geht sie bei sich Zuhause nur ungern nach unten in den **Keller*** Expected | ***Garten.*** Unexpected

*As Anne is afraid of spiders, she doesn't like going down to the **basement*** Expected | ***garden.*** Unexpected

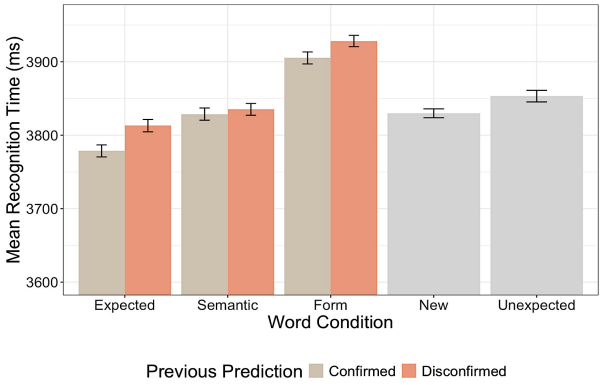
2. Implicit Memory Test

- Faster recognition times indicate pre-activations
- Conditions:
  - Keller (basement) → Expected
  - Treppe (stairs) → Semantic
  - Kellner (waiter) → Form
  - Garten (garden) → Unexpected
  - (& New Words)

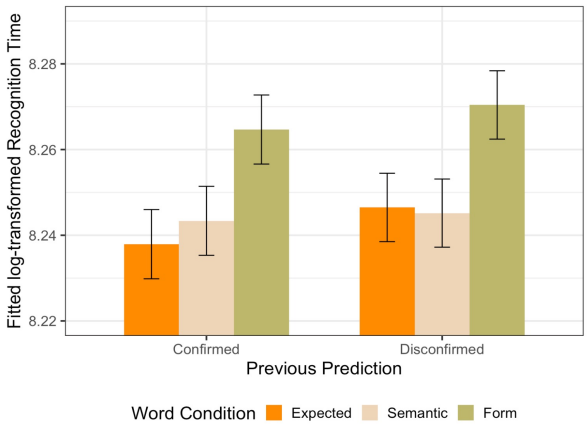
→ 77 native German-speaking adults (Mean age of 22.9 years, Range = 18-35 years, m = 24, f = 52, nb = 1)

Results

Raw Recognition Times



Fitted Recognition Times



log(recognition time in ms) ~ word condition_sum_coded * (dis)confirmed_treatment_coded + scale(trial number in implicit memory test) + scale(word frequency) + scale(word length) + scale(number of form-related neighbors) + scale(visual acuity) + scale(cloze probability of the corresponding sentence) + scale(trial number in self-paced reading task) + (1 | participants) + (1 + (dis)confirmed | words)

Discussion

- **Faster** recognition times after **prediction confirmations**, i.e., lingered pre-activations of expected words and their related neighbors in memory
- **Slower** recognition times after **prediction disconfirmations**, i.e., weakening of their memory representations to ensure successful processing of the unexpected word
- **Suppression** of **form-related neighbors** regardless of confirmation/disconfirmation [7]

But:

- **No recognition time advantage for expected words** over their semantic neighbors
- Activation of expected and semantic words not necessarily based on prediction, but possibly because they are associated with the described event in the sentences and were therefore co-activated (i.e., activation of event knowledge) [8]

→ **Inconclusive findings** of true downstream prediction effect

[1] Federmeier, K. D., & Kutas, M. (1999). A rose by any other name: long-term memory structure and sentence processing. *Journal of Memory and Language*, *41*(4), 469-495. https://doi.org/10.1006/jmla.1999.2660 [2] Huettig, F., Audring, J., & Jackendoff, R. (2022). A parallel architecture perspective on pre-activation and prediction in language processing. *Cognition*, *224*, 105050. https://doi.org/10.1016/j.cognition.2022.105050 [3] Nieuwland, M. S. (2019). Do 'early' brain responses reveal word form prediction during language comprehension? A critical review. *Neuroscience & Biobehavioral Reviews*, *96*, 367-400. https://doi.org/10.1016/j.neubiorev.2018.11.019 [4] Haeuser, K. I. & Kray, J. (in press). Do you remember? Effects of lexical predictability on false and true recognition memory. *Journal of Experimental Psychology: Learning, Memory and Cognition*. [5] Hubbard, R. J., & Federmeier, K. D. (2024). The Impact of Linguistic Prediction Violations on Downstream Recognition Memory and Sentence Recall. *Journal of Cognitive Neuroscience*, *36*(1), 1-23. https://doi.org/10.1162/jocn_a_02078 [6] Hubbard, R. J., Rommers, J., Jacobs, C. L., & Federmeier, K. D. (2019). Downstream behavioral and electrophysiological consequences of word prediction on recognition memory. *Frontiers in Human Neuroscience*, *13*, 291. https://doi.org/10.3389/fnhum.2019.00291 [7] Haeuser, K. I., & Borovsky, A. (2024). Predictive processing suppresses form-related words with overlapping onsets. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *46*. https://escholarship.org/uc/item/95w210ck [8] Metusalem, R., Kutas, M., Urbach, T. P., Hare, M., McRae, K., & Elman, J. L. (2012). Generalized event knowledge activation during online sentence comprehension. *Journal of Memory and Language*, *66*(4), 545-567. https://doi.org/10.1016/j.jml.2012.01.001

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